

Description of software system

“ALAMEL”

software and manual

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ALAMEL software is based on the Advanced LAMEL model developed by Paul Van Houtte at the department of MTM, KU Leuven. ALAMEL model is a Taylor based Micromechanical model aimed at simulating crystallographic textures and plastic anisotropy resulting from plastic deformation of cubic metals. In ALAMEL model, a microstructure is described by a set of grain boundary segments with associated grain boundary zones (GBZ). As in LAMEL model, the average shape change of a set of two GBZ's is equated to the average shape change of the polycrystal, but in each of the two zones shears are allowed to take place, which are referred to as *relaxations*. Detailed description on the model can be found in Reference 1.

The current version of the software is compiled using Compaq Visual Fortran Version 6.5 and has been tested on PC's with XP, 2000 and NT professional Windows operating system.

1. Purpose

ALAMEL essentially simulates the effect of one or more small increments of plastic strain on a polycrystalline material. So it needs:

- a description of the possible slip systems of the material (twinning systems have not yet been tested at present). These descriptions are implemented in files with suffix .PRE, made by the program PRETAY on the basis of hand-made lists of potential slip systems.
- a file describing the crystallographic texture of the material. The texture is not represented as a continuous ODF-function (using *C*-coefficients), but as a discrete

set of crystallite orientations, each with a weight (number proportional to volume fraction). Two types of such files can be used:

- 1) a file with suffix .SMT, made by hand or generated by the program ODFLAM from the MTMTAY/MTM-FHM software system. This program converts a continuous ODF-function as obtained from *C*-coefficients to a discrete set of crystallite orientations¹
 - 2) a file with suffix .CUR (“Current condition of the texture”), generated during a previous deformation texture simulation of ALamel itself .
- A description of the strain increments to be simulated (in the parameter file .I01).

ALAMEL-programs generate .CUR-output files with “Current Conditions of the Texture” (i.e. initial, intermediate and final textures + grain shapes), .RES-output files with plastic stresses, Taylor factors, rate of plastic work etc. associated to each strain increment (and for each grain), and .LST-output files with general logging information, average stresses etc.

At present, there is a program “MAIN1” for the simulation of deformation textures (many strain increments, or “deformation steps”, one after the other).

2. Organisation of the Software

2.1 Preparation of a Simulation

Use the program PRETAY to generate a data set with suffix .PRE, which contains the description of the slip systems to be considered during the simulation. PRETAY itself uses a list of slip systems which is easy to make. In fact, PRETAY already performs certain calculations which only have to be performed once.

There are already a number of .PRE-files available:

¹ In case of grain interaction models such as LAMEL model, the initial texture description should not possess any kind of sample symmetry (TRICLINIC / CLASS F in MTM FHM system should be used.)

- FCC.PRE, for 12 $\{111\}\langle 110 \rangle$ -slip systems;
 - FCCT.PRE, for 12 $\{111\}\langle 110 \rangle$ -slip systems + 12 $\{111\}\langle 112 \rangle$ twinning systems (not yet tested)
 - BCC.PRE, for 12 $\{110\}\langle 111 \rangle$ + 12 $\{112\}\langle 111 \rangle$ slip systems;
 - BCC2.PRE, for 12 $\{110\}\langle 111 \rangle$ + 12 $\{112\}\langle 111 \rangle$ + 24 $\{123\}\langle 111 \rangle$ slip systems.
- Use PRETAY only when you come up with yet another set of slip system families. The maximum number of slip systems is 48.

2.2 Simulations

The software which actually performs the simulations is a vast set of subroutines organised in the files ALG0.FOR, ALG1.FOR, ALG2.FOR, SLIPRAT.FOR, and PANCAKE2.FOR. In principle these files are not changed for the various applications. These subroutines are called from a main program. Several versions exist, and new versions can be made by the users. These main programs take care of:

- reading of certain parameters, and of the CRSSR values (critical resolved shear stress ratios);
- opening most files;
- calling the LEESOR and the SIMUL subroutines of the general package, in any order which seems appropriate.

It is necessary to edit an input parameter file such as MAIN1.I01, in which the parameters of the main program and of LEESOR and SIMUL are provided. The name of the input parameter file is read from a control file by name Main1.CTL

The LEESOR subroutine reads the file with the input texture (a .SMT-file made by hand or made by ODFTAY, or a .CUR-file made by NewSimtex itself). LEESOR wants to read 2 lines in the parameter file .I01. Before LEESOR returns control, it rewinds the input file, which can be used again in the next call of LEESOR.

SIMUL is used in a special way. First, an “initialisation call” is to be done (and not repeated later). The first argument is then 0. SIMUL will then read its parameters in the .I01-file. After it has initialised itself, SIMUL gives control back to the main program.

Before a “real” call of SIMUL are done, the main program must read (or otherwise provide) the value of the displacement gradient DG which describes the macroscopic strain in one increment. If then SIMUL is called (with the first argument set to 1), it will organise the simulation of a certain number (*nstep*) of strain increments (as specified in the .I01 parameter file). Such a series of *nstep* deformation steps is called a “block”. SIMUL will only write the initial texture (texture generated by the previous block) to the .CUR file with the “current condition of the texture). To the “results”-file (suffix .RES), it will write the stress, M-values etc. that it finds during the first strain increment of the block.

A certain number of such calls of SIMUL are done. In the case of the MAIN1-program, a last call of SIMUL (with the first argument set to 2) is done before stopping, not for doing any simulation, just for outputting the final texture to the .CUR-file.

Example 1: MAIN1 does only 1 call of LEESOR (in the beginning) + the initialisation call of SIMUL. It could for example do 4 simulation calls of SIMUL. Each could organise 12 deformation steps (increments). Before stopping, the final call of SIMUL outputs the final texture. In this case

- 48 deformation steps are simulated;
- The initial texture + the textures after steps 12, 24, 36, 48 are put on the .CUR-file;
- The results of the 1st, 13th, 25th, 37th steps are put in the .RES file.
- The displacement gradient tensor can be set before step 1, has to remain unchanged until step 12, is set again (the same one or a different one) before step 13, has to remain unchanged until step 24, etc.

Example 2: MAIN1 could also do 48 simulation calls of SIMUL. Each could organise 1 deformation steps (increments). Before stopping, the final call of SIMUL outputs the final texture. In this case

- 48 deformation steps are simulated;
- The initial texture + the textures after all steps are put on the .CUR-file, *unless* a special parameter (which is available) is used to suppress some of this output;
- The results of the all steps are put in the .RES file, *unless* a special parameter (which is available) is used to suppress some of this output;

- The displacement gradient tensor can be set or re-set before each step.

2.3 Exploitation of the results

The program FCURODF can be used to extract a texture from a .CUR-file and convert it to C-coefficients. It is similar to the SMTODF from the MTMTAY-package.

3. Short Manual

3.1 ALAMEL

3.1.1 MAIN1

Purpose: see examples 1 and 2 in section 2.2.

Run the program by giving the command:

Main1

The program produces a log file with general information, average stress etc.: *name.LST*
Other output files are: *name.RES* (with results per step and per crystallite, see Appendix 1) and *name.CUR* (with the initial texture + a series of simulated deformation textures).

Example of the file MAIN1.I01:

Line nr. Line						
a1	testjob		Name of output files (give no extension)			
a2	fcc.pre		Slip system file			
a3	8	No. of lines with tau-crit values (Stored in FK1):				
a4	1.0	1.0	1.0	1.0	1.0	1.0
a5	1.0	1.0	1.0	1.0	1.0	1.0
a6	1.0	1.0	1.0	1.0	1.0	1.0
a7	1.0	1.0	1.0	1.0	1.0	1.0
a8	1.0	1.0	1.0	1.0	1.0	1.0
a9	1.0	1.0	1.0	1.0	1.0	1.0
a10	1.0	1.0	1.0	1.0	1.0	1.0
a11	1.0	1.0	1.0	1.0	1.0	1.0
a12	1	(Main1) NBLOC				
a13	micro1.smt		NAME OF MICROSTRUCTURE FILE			

```

b1  1      (SIMUL) NLIST (Make an output listing 0 or 1)
b2  1      (SIMUL) NFILE (Make output files 0 or 1)
b3  0      (SIMUL) NFILTW (Make output files 0 or 1)
b4  1      (SIMUL) NTEN (Print distortion tensor 0 or 1)
b5  1      (SIMUL) NSYM If =0: TAUC are set to 1; if=1: values from FK1 used.
b6  0      (SIMUL) IGLIJ 0 or 1 (a print switch. Only for very short runs!)
b7  0      (SIMUL) IPR 0-3 Print switch. All except Van Houtte must use 0
c1  3.0          Eta-Factor: Stress concentration fact. on non-deforming particle
c2  0.5          Attenuation factor (on the stress concentration)
c3  1.0          0.0          0.0          F_Microstructure
c4  0.0          1.0          0.0          F_Microstructure
c5  0.0          0.0          1.0          F_Microstructure
c6  testjob-Alame1
c7  1.486        2.476        8.357        VOCE TAU-III-1, TAU-III-S, T-IV-S
c8  2.75         0.55         VOCE THETA-1, THETA-T
d1  1      (Leesor) Type of data set for input texture (1 for SMT-file)
d2  alum39.smt          NAME OF INPUT TEXTURE FILE
d3  1      (Leesor) Chosen Block (in input data set)
e1  0      (Main) If =1: output for this block is required  INITIALISATION OF SG0
e2  0.025        0.0          0.0          Displacement gradient for this block
e3  0.0          0.0          0.0
e4  0.0          0.0          -0.025
e5  1      (SIMUL) NUMBER OF SIMULATION STEPS PER CALL      (INITIALISATION OF SG0)
e6  0  1(SIMUL) 1: relaxation allowed, for relx 1 and 2 (INITIALISATION OF SG0)
f1  1      (Leesor) Type of data set for input texture (1 for SMT-file)
f2  alum39.smt          NAME OF INPUT TEXTURE FILE
f3  1      (Leesor) Chosen Block (in input data set)
g1  1      (Main) If =1: output for this block is required.
g2  0.025        0.0          0.0          Displacement gradient for this block
g3  0.0          0.0          0.0
g4  0.0          0.0          -0.025
g5  10      (SIMUL) NUMBER OF SIMULATION STEPS PER CALL      (This is for a true simulation)
g6  1  1(SIMUL) 1: relaxation allowed, for relx 1 and 2 (This is for a true simulation)
h1  1      (SIMUL) NUMBER OF SIMUL. STEPS PER CALL (Fake call of SIMUL - for output only)
h2  0  0(SIMUL) 1: relaxation allowed, for relx 1 and 2(Fake call of SIMUL - for
output only)

```

Comments:

Many lines are self-explanatory. Fortran formats used to read these lines: usually I5 for integers, F10.0 (number of digits after decimal point is free) for numbers with decimal point, A8 for character variables.

a1: Contains *name* (max. 8 characters), will be used in the names of the output files *name.LST*, *name.RES* and *name.CUR*.

a2: name of the file containing the description of slips systems to be considered during the simulation (*eg.* fcc.pre, bcc.pre)

- a3-a11*: Lines with values of TAUCR (critical resolved shear stress ratio) for 24 slip systems. The lines are stored in the array FK1. The first 24 values: for the slip in forward direction on the 24 slip systems. The next 24 values: for slip in backward direction on the 24 slip systems. Up to 96 values can be given (6 per line).
- a12*: *nbloc*= number of times that SIMUL is called for the simulation of a “block” of deformation increments.
- a13*: name of the file containing the microstructure, as characterised by the grain boundary segments orientation distribution function (GBSODF). The file contains the Euler angles of the grain boundary segments. The provided “micro.smt” contains a uniform GBSODF as obtained from an equiaxed microstructure. For further details please refer to [1].
- b1-b4*: See comment in line.
- b5*: See comment in line. See comment on *a3-a11* for meaning of FK1.
- b6-b7*: Ask Prof. Van Houtte’s advice if you want to change this.
- c1-c5*: Ask Prof. Van Houtte’s advice if you want to change this.
- c6*: Title of the new simulation (Max. 40 characters)
- c7-c8*: Parameters describing the work hardening of the material by a simple voce type law. Please refer to figure1 for further explanation of the parameters.
- d1*: Allowed values: 1 or 2. “1”: the initial texture is in a .SMT-file; “2”: the initial texture is in a .CUR-file.
- d2*: Name of the input texture file containing discrete crystal orientations.
- d3*: Is not used if a .SMT-file is used. If a .CUR-file is used: should be the sequence number of the texture in the .CUR file which is to be taken as initial texture. Note that the first texture is the file has the sequence number 0 (it was the initial texture of the simulation which generated the .CUR-file), the 1 is the second texture in the file (i.e., deformation texture obtained after the completion of all simulations of block 1), etc. If the .CUR file contains the results of the simulation of *n* consecutive blocks of deformation increments, then the sequence number of the final texture will be *n*.
- e1-e6*: The lines are for initialization purposes, self explanatory
- f1-f3* : same as lines d1-d3

Following lines are to be repeated for each block to be simulated:


```

a7      LMAX      22  0.00
a8      testjob 100% rolling

```

Comments:

a1: See table 1 of the manual of “MTMFHM” for IMAG and IDN. For cubic metals, IDM is 4.

a2: Number of extraction runs to be performed. As an example, in a simulation with 4 blocs, if one needs the texture only after the 4th block, then the extractions performed are 1. If textures need to be extracted, say after the 2nd and the final block then the number of extractions performed is 2.

Following lines are to be repeated for each extraction run:

a3: Number of the block after which the texture is to be extracted. If you need to extract texture after n th-block, then the entry in this line should be $n+1$.

a4: "22" is the maximum degree of the series expansion (leave it as it is); "7.0" is Φ (see section 5.9 of the manual of "MTMFHM"), the spread of the Gaussian.

a5: Name for the texture.

References

1. Van Houtte, P., Li, S., Seefeldt, M., Delannay, L., 2005. “Deformation texture prediction: from the Taylor model to the Advanced Lamel model.” International Journal of Plasticity, Volume 21, Issue 3, March 2005, Pages 589–624
2. Li, S., Van Houtte, P., Engler, O., 2005. “Plastic anisotropy and texture evolution during tensile testing of extruded aluminium profiles”. modelling and simulation in materials science and engineering, Volume 13, Pages 783-795

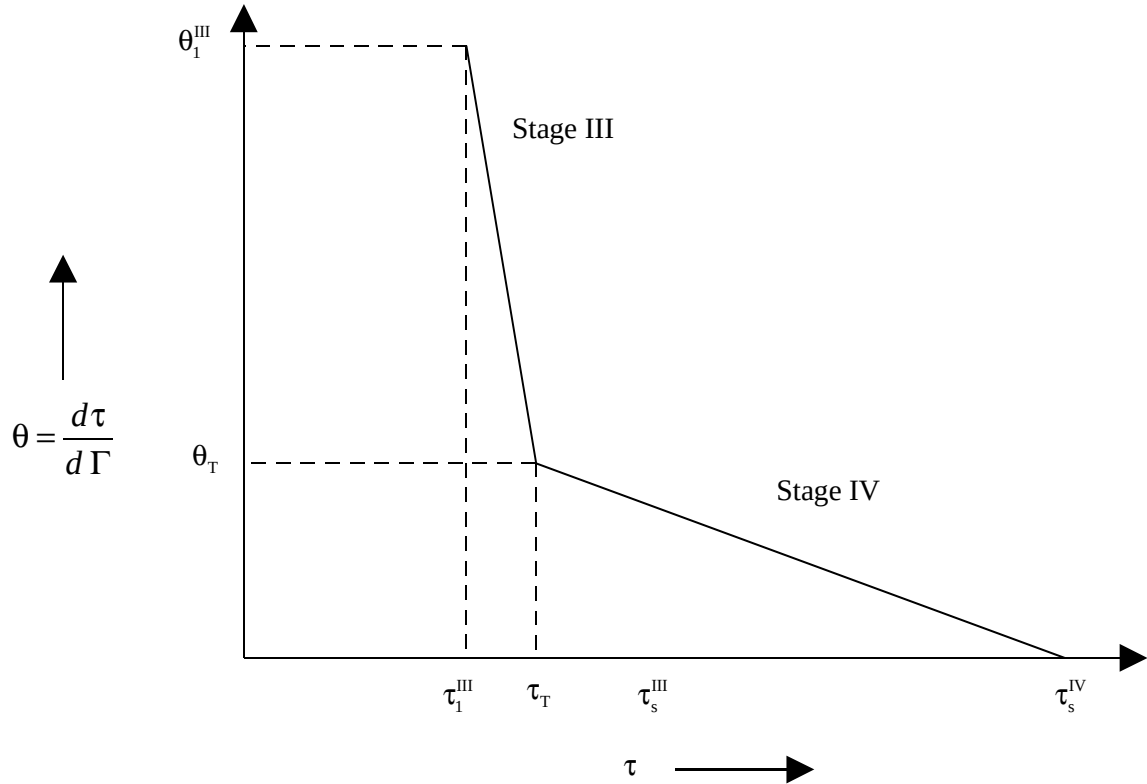


Figure 1: Schematic of a work hardening curve indicating the parameters used in such a way that a tensile curve could be reproduced by a simulation using this hardening law in a Taylor model.

Appendix 1: Description of Output File “Results”

The “Results”-file has a suffix .RES. It contains results about the response of the material to the imposed strain *during* a step of plastic deformation. A typical result is the stress. The Fortran data set reference number used in WRITE instructions is IMP2.

The simulation is subdivided in *blocks* of deformation steps. One such block is a series of deformation steps organised by a single call of the subroutine SIMUL of the file ALG0.FOR. This subroutine will produce output about the stresses etc. for the first step of a block only. So if 4 blocks of 12 steps each are simulated, one will find data about the 1st, 13th, 25th and 37th deformation step in the file (name).RES. It is an ASCII-file. Each line will be described below. It will each time be indicated where the WRITE-command is to be found (*file/subroutine/label of FORMAT-statement*).

1. General Heading:

One line with a title (ALG0/SIMUL/ 98)

Contains TITEL = the title of the simulation.

2. For each “Block”:

a) Line with sequence number of first deformation step of the block (ALG0/SIMUL/404):

Position	Name of Variable	Meaning
1-11	-	“Deformation Step Nr.”
12-16	<i>nrstep+1</i>	Sequence number of the first deformation step of the block
19-40	-	“Number of Orientations”
41-45	<i>npoint</i>	Number of crystallites in the discrete texture file

b) Line with headings (ALG0/SIMUL/404)

c) Line with 220 asterisks (ALG0/SIMUL/404)

d) for each of the *npoint* crystallites, a line with results (ALG1/TAYLR1/150):

Position	Name of Variable	Meaning
1-5	<i>ior</i>	Sequence number of crystallite
6-15	<i>wdot</i>	Rate of plastic work in the grain. i.e sum of the shears normalised by the von Mises equivalent strain rate
16-25	<i>Wdot1</i>	Equivalent flow stress of the grain normalised by the critical resolved shear stress.
26-35	<i>tau</i>	Critical resolved shear stress of the slip systems.
36-45	<i>Wdot1/tau</i>	Taylor factor
46-55	<i>ratlon</i>	$r_L = \left(\frac{\mathbf{D} + \mathbf{p}^s}{\ \mathbf{D}\ } \right) : \left(\frac{\mathbf{D}}{\ \mathbf{D}\ } \right)$; is called “the ratio of the parallel strain rates”; is the ratio of the projection of $\mathbf{D} + \mathbf{p}^s$ on the direction

		of the prescribed strain rate $\mathbf{D}$, to the length of $\mathbf{D}$. Is a measure for the magnitude of the parallel relaxation. Would be 1 if the parallel relaxation is zero, >1 if it is positive, <1 if it is negative.
56-60	-	spaces

61-70	$rhossa(1,1)$	$\mathbf{p}^s$ (symmetric part of the relaxation tensor, expressed in sample reference system)
71-80	$rhossa(2,2)$	
81-90	$rhossa(3,3)$	
91-100	$rhossa(2,3)$	
101-110	$rhossa(3,1)$	
111-120	$rhossa(1,2)$	
121-125	-	spaces
126-135	$rhoasa(2,3)$	$\mathbf{p}^a$ (antisymmetric part of the relaxation tensor, expressed in sample reference system)
136-145	$rhoasa(3,1)$	
146-155	$rhoasa(1,2)$	
156-160	-	
161-170	$ssam(1,1)$	Stress, expressed in sample reference system.
171-180	$ssam(2,2)$	
181-190	$ssam(3,3)$	
191-200	$ssam(2,3)$	
201-210	$ssam(3,1)$	
211-220	$ssam(1,2)$	

3.Final Remark

If there are *nbloc* blocks and *nstep* simulation steps in total, then at the end of the file, there will be 2 **useless** last lines with the mentioning “Record nr.” *nstep*+1.

Appendix 2: Description of Output File “Current Condition”

The “Current Condition”-file has a suffix .CUR. It contains the description of the texture and the grain morphologies at selected moments in time. The Fortran data set reference number used in WRITE instructions is IMP1.

The simulation is subdivided in *blocks* of deformation steps. One such block is a series of deformation steps organised by a single call of the subroutine SIMUL of the file ALG0.FOR. Directly after being called, this subroutine will produce output about the current condition of the texture, of the morphology of the grains, and of the total deformation. This then is the initial texture/condition of the simulations performed in the block. So if 4 blocks of 12 steps each are simulated, one will find data about the texture/condition just before the 1st simulation steep, i.e. “after step 0”. The

texture/condition after deformation steps 12th, 24th and 36th step will then be outputted. Then a last call of SIMUL is done, without any simulation of deformation, just for the sake of outputting the final texture/condition (after step 48) in the .CUR-file.

The .CUR-file is an ASCII-file. Each line will be described below. It will each time be indicated where the WRITE-command is to be found (*file/subroutine/label of FORMAT-statement*).

1. General Heading:

One line with a title (ALG0/SIMUL/ 98)

Contains TITEL = the title of the simulation.

2. For each “Block” (including the fictitious last one):

a) Blank line (ALG0/SIMUL/402)

b) Line with headings (ALG0/SIMUL/402)

c) Line with some general information (ALG0/SIMUL/403)

Position	Name of Variable	Meaning
1-6	<i>nrstep</i>	Nr. of last deformation step which has been performed
7-11	-	spaces
12-16	<i>npoin</i> t	Number of crystallites in the discrete texture file
17-62	-	spaces
63-72	<i>F</i> (1,1)	First column of the macroscopic deformation gradient tensor
73-82	<i>F</i> (2,1)	
83-92	<i>F</i> (3,1)	
93-94	-	spaces
95-104	<i>F</i> (1,2)	Second column of the macroscopic deformation gradient tensor
105-114	<i>F</i> (2,2)	
115-124	<i>F</i> (3,2)	
125-126	-	spaces
127-136	<i>F</i> (1,3)	Third column of the macroscopic deformation gradient tensor
137-146	<i>F</i> (2,3)	

147-156	$F(3,3)$	
157-158	-	spaces
159-168	$GAXES(1)$	Half - axes a, b, c of “average grain” (ellipsoid)
169-178	$GAXES(2)$	
179-188	$GAXES(3)$	
189-190	-	spaces
191-200	$GLR(1)$	Morphological Euler angle φ_1
201-210	$GLR(2)$	Morphological Euler angle Φ
211-220	$GLR(3)$	Morphological Euler angle φ_2

The “morphological Euler angles” describe the orientation of the orthogonal axes system formed by the 3 principal axes of the ellipsoid which we use as model for the “average grain”.

d) Line with headings (ALG0/SIMUL/401)

e) For each crystallite:

Line with orientation of the crystallite, and other information (ALG0/SIMUL/400)

Position	Name of Variable	Meaning
1-6	Ior	Sequence number of crystallite
7-16	$Gewf$	Weighting factor of crystallite
17-18	-	spaces
19-28	$\tilde{f}1$	Euler angle φ_1 (for crystallographic orientation)
29-38	Φ	Euler angle Φ
39-48	$\tilde{f}2$	Euler angle φ_2
49-50	-	spaces

51-60	$Gmma$	$\Gamma = \sum_{i=1}^{nrstep} M_{RC}$ Sum of all the M_{RC} -values since the simulation started (see Appendix 1, 2d, position 6-15)
61-62	-	spaces
63-72	$F(1,1)$	First column of the local deformation gradient tensor
73-82	$F(2,1)$	
83-92	$F(3,1)$	
93-94	-	spaces
95-104	$F(1,2)$	Second column of the local deformation gradient tensor
105-114	$F(2,2)$	
115-124	$F(3,2)$	
125-126	-	spaces

127-136	$F(1,3)$	Third column of the local deformation gradient tensor
137-146	$F(2,3)$	
147-156	$F(3,3)$	
157-158	-	spaces
159-168	$GAXES(1)$	Half - axes a, b, c of the grain (ellipsoid)
169-178	$GAXES(2)$	
179-188	$GAXES(3)$	
189-190	-	spaces
191-200	$GLR(1)$	Morphological Euler angle ϕ_1
201-210	$GLR(2)$	Morphological Euler angle Φ
211-220	$GLR(3)$	Morphological Euler angle ϕ_2

Appendix 3: Calculation of planar anisotropy (r-value) of sheets using ALAMEL software system

The contraction ratio $r = \frac{D_{22}}{D_{33}}$ is often used to characterise the formability of metallic sheets. The r -values can be obtained both by experiments and simulations (More discussion can be found in Reference 2). Here a procedure to obtain the necessary values using the current software is briefly described:

The tensile test is a stress based test where in the stress tensor has only one nonzero component along the direction of elongation. However, the current software takes strain mode as an input. To perform a tensile test, a general strain mode tensor is used, which looks like a strain rate tensor corresponding to elongation in the direction x_1 , but with a unit strain rate in the direction x_1 :

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -q & 0 \\ 0 & 0 & -(1-q) \end{bmatrix} \quad \text{where } q = \frac{r}{1+r}$$

A given texture is subjected to a single step of very small tensile strain (along direction 1) as defined above, for values of q ranging from 0 to 1. For each value of q , the stress component along the elongation direction (σ_{11}) is calculated. Variation of σ_{11} with q is the plotted; it can be observed that σ_{11} reaches a minimum for a particular value of q , say q^* . r -value is then calculated using the formula $r = \frac{q^*}{1-q^*}$. It can be shown that at q^* ,

σ_{11} is the only nonzero stress component, hence corresponding to a real tensile test.

σ_{11} can be obtained from $wdot$, which is printed in the *.RES output file. It represents the rate of plastic work dissipated per unit volume, but normalised by the von Mises equivalent strain rate D_{eq} . Note that the plastic work dissipated per unit volume and per unit time is equal to the product ($wdot$) D_{eq} .

D_{eq} , the von Mises equivalent strain rate, is defined as $\sqrt{\frac{2}{3} D_{ij} D_{ij}}$. It is a function of q .

D_{11} is the strain rate in the elongation direction.

The stress in longitudinal direction is then:

$$\sigma_{11} = w \dot{\epsilon} \frac{D_{eq}}{D_{11}}$$

This finally leads to:

$$\sigma_{11} = 2 w \dot{\epsilon} \sqrt{\frac{1 - q + q^2}{3}}$$

of which, as said above, the minimum with respect to q must be sought.